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UNIT I

Experiment 1: Basic input and output using different data types

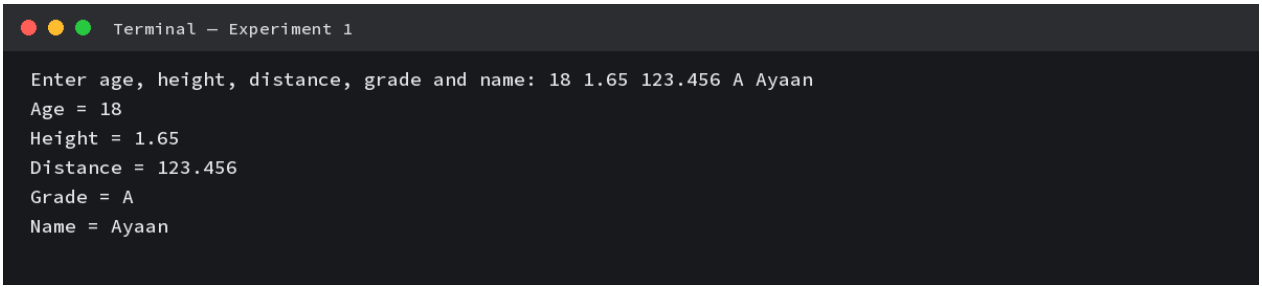
```
#include <stdio.h>

int main(void) {
    int age;
    float height;
    double distance;
    char grade;
    char name[30];

    printf("Enter age, height, distance, grade and name: ");
    scanf("%d %f %lf %c %29s", &age, &height,
          &distance, &grade, name);

    printf("Age = %d\n", age);
    printf("Height = %.2f\n", height);
    printf("Distance = %.3f\n", distance);
    printf("Grade = %c\n", grade);
    printf("Name = %s\n", name);
    return 0;
}
```

Output:



UNIT I

Experiment 2: Calculate the area of a circle

```
#include <stdio.h>
#define PI 3.14159

int main(void) {
    float radius, area;

    printf("Enter the radius of the circle: ");
    scanf("%f", &radius);
    area = PI * radius * radius;
    printf("Area of the circle = %.2f\n", area);
    return 0;
}
```

Output:



UNIT I

Experiment 3: Calculate the average of three numbers

```c
#include <stdio.h>

int main(void) {
    float a, b, c, average;

    printf("Enter three numbers: ");
    scanf("%f %f %f", &a, &b, &c);
    average = (a + b + c) / 3.0f;
    printf("Average of the three numbers = %.2f\n", average);
    return 0;
}
```

Output:



UNIT II

Experiment 6: Demonstrate operator precedence and associativity

```c
#include <stdio.h>

int main(void) {
    int a = 5, b = 10, c = 2;
    int result1 = a + b * c;
    int result2 = (a + b) * c;
    int x = 5, y = 10, z = 15;
    int logicResult = x < y && y < z;

    printf("a + b * c       = %d\n", result1);
    printf("(a + b) * c     = %d\n", result2);
    printf("x < y && y < z  = %d\n", logicResult);
    return 0;
}
```

Output:



UNIT II

Experiment 7: Check whether a number is even or odd

```c
#include <stdio.h>

int main(void) {
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);
    if (number % 2 == 0) {
        printf("%d is Even\n", number);
    } else {
        printf("%d is Odd\n", number);
    }
    return 0;
}
```

Output:



UNIT II

Experiment 8: Determine whether a number is positive, negative, or zero

```c
#include <stdio.h>

int main(void) {
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);
    if (number > 0) {
        printf("%d is Positive\n", number);
    } else if (number < 0) {
        printf("%d is Negative\n", number);
    } else {
        printf("The number is Zero\n");
    }
    return 0;
}
```

Output:



UNIT III

Experiment 12: Find the sum of elements in a one-dimensional array

```
#include <stdio.h>

int main(void) {
    int arr[100], n, i, sum = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        sum += arr[i];
    }
    printf("Sum of the array elements = %d\n", sum);
    return 0;
}
```

Output:



UNIT III

Experiment 13: Add two matrices of the same order

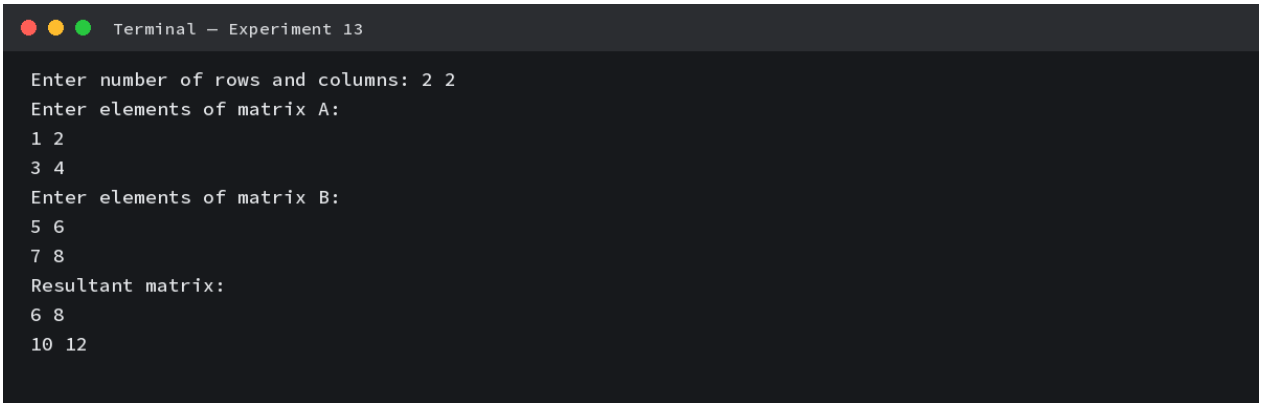
```c
#include <stdio.h>

int main(void) {
    int a[10][10], b[10][10], sum[10][10];
    int rows, cols, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &rows, &cols);
    printf("Enter elements of matrix A:\n");
    for (i = 0; i < rows; i++)
        for (j = 0; j < cols; j++) scanf("%d", &a[i][j]);
    printf("Enter elements of matrix B:\n");
    for (i = 0; i < rows; i++)
        for (j = 0; j < cols; j++) scanf("%d", &b[i][j]);

    printf("Resultant matrix:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            sum[i][j] = a[i][j] + b[i][j];
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output:



UNIT III

Experiment 14: Find factorial using a user-defined function

```
#include <stdio.h>

long factorial(int n);

int main(void) {
    int num;
    long result;

    printf("Enter a number: ");
    scanf("%d", &num);
    result = factorial(num);
    printf("Factorial of %d = %ld\n", num, result);
    return 0;
}

long factorial(int n) {
    long fact = 1;
    int i;
    for (i = 1; i <= n; i++) {
        fact *= i;
    }
    return fact;
}
```

Output:

